

Regulations

Goal: Apply the FAA regulations to real-world scenarios. Determine if each scenario is compliant. If not compliant, suggest changes to bring it into compliance.

1. Your friend is a realtor in town and has a company that usually does the drone photography for her houses but the company can't fly a house because they are booked up for the month. She asks you if you can do it with your drone. You don't have your part 107 certificate, but your drone is a DJI mini and weighs less than .55 lbs. Your friend offers you gift cards to Target instead of a cash payment as you are not certified. Would this operation fall under part 107? Is this flight OK?
2. You are flying a night flight over an antenna array to detect "hot spots". You have your remote pilot certificate and you have a high intensity light that can be seen 3 miles away on your drone. There are houses nearby and a homeowner has stated that the lights are keeping them awake at night. You offer to turn down the brightness and they agree that this is a good compromise.
3. You are flying a large 45 lbs. fixed wing drone. You are having some weight and balance issues. You need to add some nose weight but don't have anything except some lithium batteries. They are the same batteries that the drone uses and fit perfectly and are just the right weight to balance out the drone. You are not using the power from the batteries but they are safely enclosed in the aircraft, they are of the same make and model that the airplane uses anyway. Are there any issues with doing this?
4. You have two new drones. They are very easy to fly and can take off, fly and land on their own. You are doing field surveys with the drones. You literally just need to push a button and they take off, fly their grid and land. You decide to fly them both at the same time. You also have two helpers with you that can help monitor the drone. You have your remote pilot certificate but they don't. All three of you are monitoring the drones and can easily take over at any time. Is this ok?
5. Your drone's instructor is doing a drone demo and it goes horribly wrong. The drone goes out of control and hits several students. They all get cut up pretty bad but are bandaged up on site with the class first aid kit. No one needed to go to the hospital, but a few bleed quite a bit before being patched up. Do you need to report this to the FAA? If so, within how many days?
6. You are doing a livestock survey for a local rancher. You have your remote pilot certificate. Your drone experiences a malfunction and hits a brand new John deer tractor and breaks the glass. The cost to repair the glass is \$450. Do you need to report this to the FAA? If so, within how many days?
7. You have your remote pilot certificate. Your best friend wants to see what your job is like so you take him along. He wants to fly the drone on today's job. You allow him to do so, but you stay right next to him and you feel that you can grab the controls and take over if anything went wrong. Nothing bad happens and the flights for the day are all successful. Is this ok?

8. Your daughter is 12 years old. You buy her a drone for Christmas. It weighs 2 lbs. so you register it for her with the FAA. She flies it in your backyard. She gets pretty good at it so you leave her by herself to fly unsupervised. Is this ok?
9. You moved out of your apartment in May. It is now September. You only moved across town so you still live in Montana, in fact, still in Bozeman. You have your part 107 certificate. Your drone company films backcountry skiers. They pay you to get footage that they use on their Youtube channels. You have not yet updated your address with the FAA. You plan to do it next week. Today is a powder day and you need to go film them skiing. Is this ok?
10. You want to film the northern lights with your drone. You have a camera that can take still pictures between the flashes of your strobe light so there is no issue running the strobe. However, you want to fly up to 2,000 feet. You will need to apply for a waiver for this flight. You plan to fly next weekend. Do you have enough time to get this waiver?